

LAB EXPERIMENT RECORD

06-10-2026---Set 11: Product Category Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Register Number	192325016
Date	06 OCTOBER 2026
Day	TUESDAY

Aim

To analyze product category sales using a funnel chart, table and pie chart, and to build an interactive Tableau dashboard for exploring product category data.

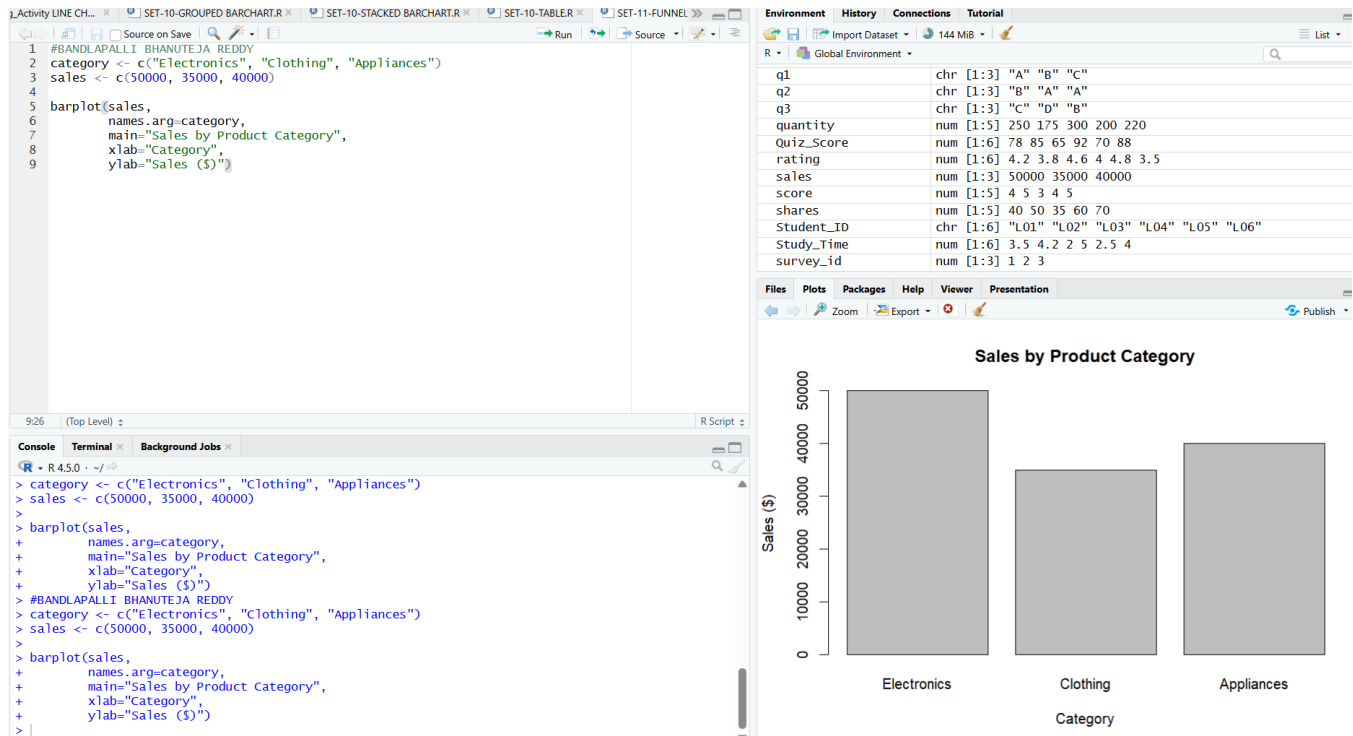
Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

Question 1: Funnel Chart of Product Categories

Question: Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

Aim: To analyze the sales of each product category as stages of a funnel using a labelled and titled funnel chart.



Result: The funnel narrows from Electronics (50000) to Appliances (40000) to Clothing (35000), so Clothing keeps 70% of the top stage; the data has no separate conversion stages, so the categories are ranked by sales.

Question 2: Table of Product Category Sales

Question: Build a table to display the sales data for each product category. Label the table elements.

Aim: To present the sales of each product category in a clearly labelled table.

The screenshot shows the RStudio IDE with four tabs at the top: '10-GROUPED BARCHART.R', 'SET-10-STACKED BARCHART.R', 'SET-10-TABLE.R', and 'SET-11-FUNNEL CHAI'. The active tab is 'SET-10-TABLE.R'. The editor window contains the following R code:

```

1 #BANDLAPALLI BHANUTEJA REDDY
2 category <- c("Electronics", "Clothing", "Appliances")
3 sales <- c(50000, 35000, 40000)
4 data <- data.frame(category, sales)
5 print(data)

```

The console window at the bottom shows the execution of the code:

```

> category <- c("Electronics", "Clothing", "Appliances")
> sales <- c(50000, 35000, 40000)
>
> barplot(sales,
+         names.arg=category,
+         main="Sales by Product Category",
+         xlab="Category",
+         ylab="Sales ($)")
> #BANDLAPALLI BHANUTEJA REDDY
> category <- c("Electronics", "Clothing", "Appliances")
> sales <- c(50000, 35000, 40000)
> data <- data.frame(category, sales)
> print(data)
  category sales
1 Electronics 50000
2  Clothing  35000
3 Appliances  40000
> |

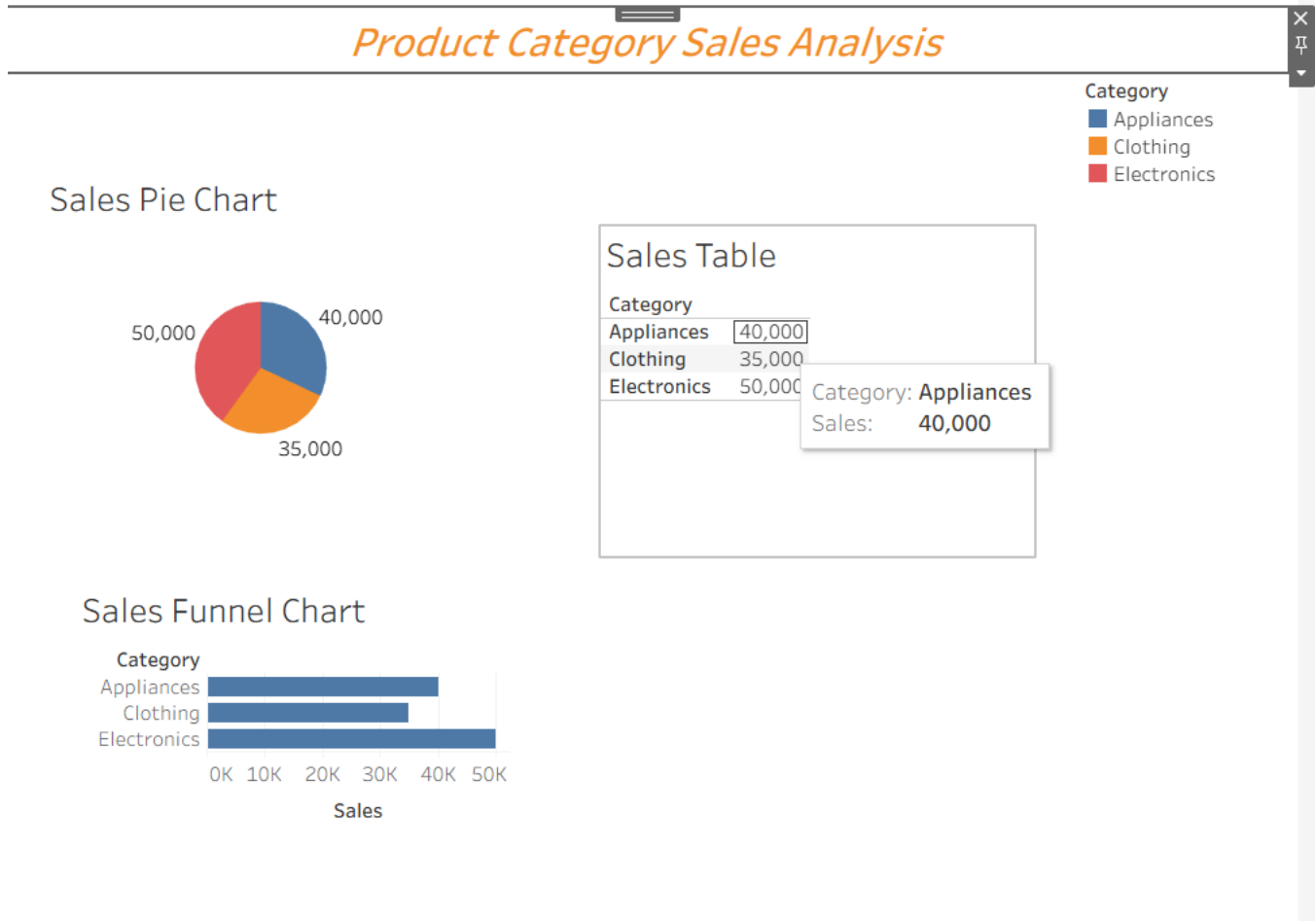
```

Result: The table lists the three categories with their sales, and the total sales are 125000.

Question 3: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

Aim: To build an interactive Tableau dashboard that combines the pie chart, funnel chart and table for exploring product category data.

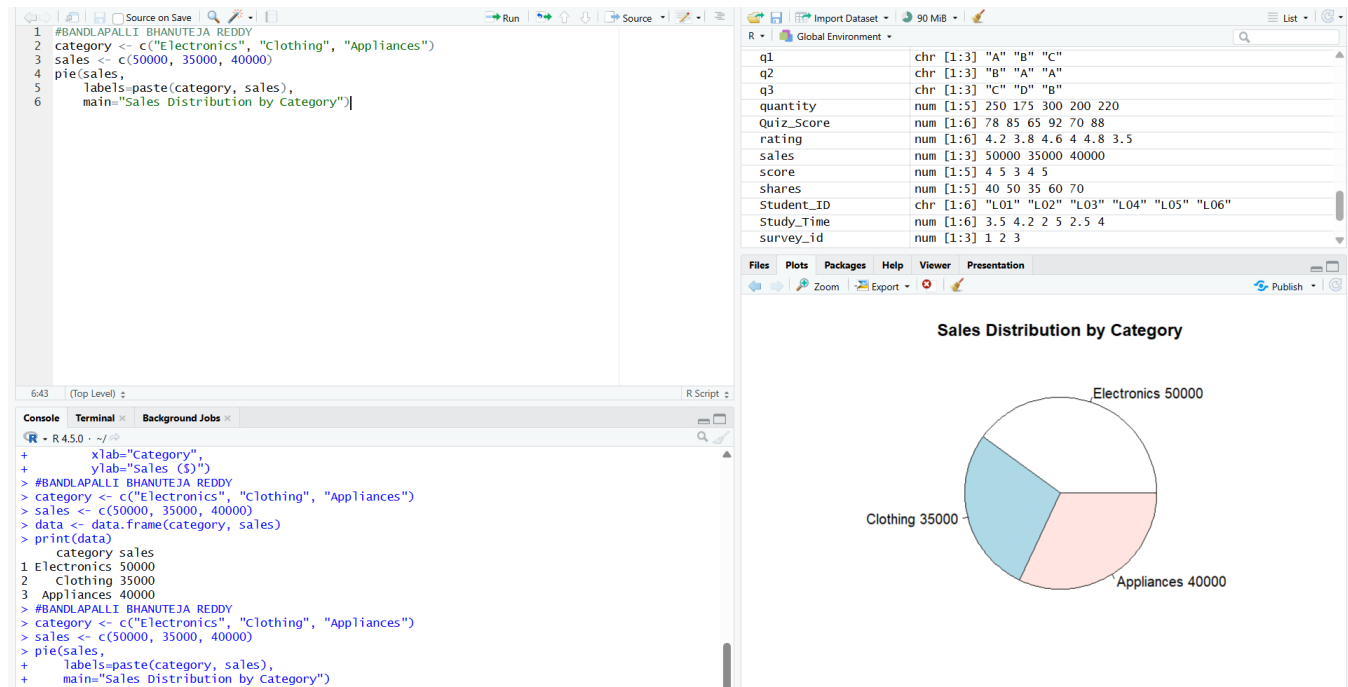


Result: The dashboard places all three views together, and a category filter updates them so users can explore the sales interactively.

Question 4: Pie Chart of Sales by Category

Question: Create a pie chart to represent the distribution of sales across product categories. Include labels.

Aim: To represent the share of sales of each product category using a labelled pie chart.



Result: Electronics holds 40% of sales, Appliances 32% and Clothing 28%.

Overall Result

Observation: Total sales are 125000, with Electronics the largest category (40%) and Clothing the smallest (28%).